Colegio de San Gabriel Arcangel, Inc.
Founded 1993
City of San Jose del Monte, Bulacan, Philippines

COBRA: COMPREHENSIVE ONLINE BEGINNER RESOURCE FOR PYTHON APPLICATION

A Capstone Project
Submitted to the Faculty of the College of Computer Studies
Colegio de San Gabriel Arcangel, Inc.

In Partial Fulfillment of the Requirements for the
Degree of Bachelor of Science in Information Technology

Submitted by:

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Darell F. Bacani
Christian Gold DC. Española
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Nicole S. Mijoy
Emmanuel A. Spicer

Prof. Jimmy D.V. Roldan
Research Group's Adviser

Submitted to:

Jimmy D.V. Roldan, MSIT
Research Course Facilitator

October 2026

CERTIFICATE OF ORIGINALITY

This is to certify that the intellectual content of this research paper entitled **"COBRA: COMPREHENSIVE ONLINE BEGINNER RESOURCE FOR PYTHON APPLICATION"** is the result of our collective ideas derived from the data gathered with the guidance of the professionals who have assisted us in the construction of this research. Even though we received help with style, presentation, and language expression from others, we utilized suitable citations in referring to other works from other sources.

This research paper does not contain any previously published or written material by another person which, to a substantial extent, has been accepted for any other degree or diploma at Colegio de San Gabriel Arcangel, Inc. or any other educational institution, except where due acknowledgment is made in the paper. Any contributions made to the research by others, with whom we have collaborated at Colegio de San Gabriel Arcangel, Inc. or elsewhere, are explicitly acknowledged.

We confirm that we understand plagiarism and accept the repercussions of any plagiarized work submitted by us, either individually or as part of a group.

Researchers:

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Christian Gold DC. Española	**Markgil L. Gamboa**
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Emmanuel A. Spicer	

Attested by:

Prof. Jimmy D.V. Roldan	**Jimmy D.V. Roldan, MSIT**
Research Group's Adviser	Research Course Facilitator

Date: _____

CERTIFICATE OF EDITING

This is to certify that I, the grammarian, have edited this thesis entitled **"COBRA: COMPREHENSIVE ONLINE BEGINNER RESOURCE FOR PYTHON APPLICATION"**

prepared by:

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Emmanuel A. Spicer

and have found it to be thorough and acceptable with respect to grammar and composition.

Name of Grammarian
Signature over printed name
Designation: _____
Date of Completion: _____

APPROVAL SHEET

This capstone project entitled **"COBRA: COMPREHENSIVE ONLINE BEGINNER RESOURCE FOR PYTHON APPLICATION"** was prepared and submitted by:

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Emmanuel A. Spicer
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Nicole S. Mijoy

in partial fulfillment of the requirements for the degree of **Bachelor of Science in Information Technology** has been examined and hereby recommended for approval and acceptance for ORAL EXAMINATION.

Prof. Jimmy D.V. Roldan	**Jimmy D.V. Roldan, MSIT**
Research Group's Adviser	Research Course Facilitator

APPROVED by the COMMITTEE ON ORAL EXAMINATION with a grade of _____.

Name of Panel Member	**Name of Panel Member**
Member	Member

Name of Panel Chairman
Chairman

Date of Oral Examination: _____

ACCEPTED and APPROVED as partial fulfillment of the requirements for the degree of **Bachelor of Science in Information Technology**.

Name of Dean
Dean, College of Computer Studies

ACKNOWLEDGMENTS

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Chapter 1

INTRODUCTION, REVIEW OF RELATED LITERATURE,

AND THE PROBLEM

[insert introductory paragraph explaining the contents of this chapter]

Introduction

Review of Related Literature

Theoretical Framework

Conceptual Framework or Research Paradigm

Statement of the Problem

Hypothesis

Scope and Delimitation of the Study

Significance of the Study

Definition of Terms

Chapter 2

METHODOLOGY

[insert introductory paragraph explaining the contents of this chapter]

Research Design

Research Locale

Respondents of the Study

Sampling Design

Instrumentation

Validation of the Instrument

Data Gathering Procedure

Ethical Considerations

Statistical Treatment of the Data

Chapter 3

RESULT, ANALYSIS, AND DISCUSSION

[insert introductory paragraph explaining the contents of this chapter]

Research Problem 1:

Research Problem 2:

Research Problem 3:

Research Problem 4:

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Chapter 4

SUMMARY OF FINDINGS, CONCLUSIONS, AND RECOMMENDATIONS

[insert introductory paragraph explaining the contents of this chapter]

Summary of Findings

Relative to Problem 1:

Relative to Problem 2:

Relative to Problem 3:

Conclusions

Relative to Finding 1:

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Relative to Finding 3:

Recommendations

Relative to Conclusion 1:

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Other Recommendations

REFERENCES

Books

Published Theses and Dissertation

Journal and Articles

Online Sources

APPENDIX A:
LETTER TO RESPONDENTS

COLEGIO DE SAN GABRIEL ARCANGEL, INC.
College of Computer Studies
City of San Jose del Monte, Bulacan

Date: _____

Sir / Madam:

Greetings of Peace and Joy!

We are the students of Colegio de San Gabriel Arcangel, Inc. conducting a study entitled **"COBRA: COMPREHENSIVE ONLINE BEGINNER RESOURCE FOR PYTHON APPLICATION,"** which aims to develop a web-based learning system that helps beginners learn Python programming. This study is part of the requirements for the degree of Bachelor of Science in Information Technology.

In this regard, we would like to request your participation in evaluating the system by using it and answering our survey questionnaire. Rest assured that the data gathered will be held with the utmost confidentiality in accordance with the Data Privacy Act of 2012.

Your favorable response will be of great help to the realization of this study.

Thank you very much.

Very truly yours,

Name of Student Lead Researcher
Student Lead Researcher

Noted by:

Prof. Jimmy D.V. Roldan	**Jimmy D.V. Roldan, MSIT**
Research Group's Adviser	Research Course Facilitator

APPENDIX B:
LETTER OF PERMISSION TO THE DEAN

COLEGIO DE SAN GABRIEL ARCANGEL, INC.
College of Computer Studies
City of San Jose del Monte, Bulacan

Date: _____

Name of Dean
Dean, College of Computer Studies
Colegio de San Gabriel Arcangel, Inc.

Sir / Madam:

Greetings of Peace and Joy!

We are the students of Colegio de San Gabriel Arcangel, Inc. conducting a study entitled **"COBRA: COMPREHENSIVE ONLINE BEGINNER RESOURCE FOR PYTHON APPLICATION,"** which aims to develop a web-based learning system that helps beginners learn Python programming. This study is part of the requirements for the degree of Bachelor of Science in Information Technology.

In this regard, we would like to request your good office **to allow us to conduct the evaluation of our system with the students and faculty of the college**. Rest assured that the data gathered will be held with the utmost confidentiality.

Your favorable response will be of great help to the realization of this study.

Thank you very much.

Very truly yours,

Name of Student Lead Researcher
Student Lead Researcher

Noted by:

Prof. Jimmy D.V. Roldan	**Jimmy D.V. Roldan, MSIT**
Research Group's Adviser	Research Course Facilitator

APPENDIX C:
LETTER TO EXPERT VALIDATORS
FOR THE VALIDATION OF THE SURVEY QUESTIONNAIRE

COLEGIO DE SAN GABRIEL ARCANGEL, INC.
College of Computer Studies
City of San Jose del Monte, Bulacan

Date: _____

Sir / Madam:

Greetings of Peace and Joy!

We are the students of Colegio de San Gabriel Arcangel, Inc. conducting a study entitled **"COBRA: COMPREHENSIVE ONLINE BEGINNER RESOURCE FOR PYTHON APPLICATION."** This study is part of the requirements for the degree of Bachelor of Science in Information Technology.

In this regard, may we have the pride and honor of having you as one of the research experts to validate our survey questionnaire by giving your invaluable comments and suggestions. Your much-sought participation will lend the necessary credence to our research work.

Your favorable attention and action on our request will significantly help the complete realization of this study. Thank you very much.

Very truly yours,

Name of Student Lead Researcher
Student Lead Researcher

Noted by:

Prof. Jimmy D.V. Roldan	**Jimmy D.V. Roldan, MSIT**
Research Group's Adviser	Research Course Facilitator

APPENDIX D:
RESEARCH INSTRUMENT VALIDATION SHEET
FOR QUANTITATIVE RESEARCH

Name of Validator: _____
Degree/Specialization: _____
Position/Designation: _____

Research Title: COBRA: COMPREHENSIVE ONLINE BEGINNER RESOURCE FOR PYTHON APPLICATION

Direction: This tool asks for your evaluation of the questionnaire to be used in data gathering for the investigation stated above to establish its validity. You are requested to give your honest assessment using the criteria stated below. Please choose only one from the selection.

Scale	Interpretation	Description
4	Highly Valid	The questionnaire is highly valid and can provide unbiased data for investigation, allowing 0–5% error.
3	Valid	The questionnaire is valid and can provide unbiased data for investigation, allowing 6–10% error.
2	Less Valid	The questionnaire is less valid, unacceptable, and needs revisions.
1	Not Valid at All	The questionnaire is not valid at all, unacceptable, and needs revisions.

Indicators	4	3	2	1
Clarity and Direction of Items				
1. The vocabulary level, language, and structure are at the conceptual level of the participants/respondents.	☐	☐	☐	☐
2. The test directions and the items are written clearly and understandably.	☐	☐	☐	☐
Presentation and Organization of Items				
3. The items are presented and organized in a logical manner.	☐	☐	☐	☐
Suitability of Items				
4. The items appropriately present the substance/content of the research.	☐	☐	☐	☐
5. The questions/items are designed to determine what is supposed to be measured.	☐	☐	☐	☐
Adequateness of the Content				
6. The number of questions/items per domain/area is representative enough of all the questions needed for the research.	☐	☐	☐	☐
Attainment of the Purpose				
7. The instrument as a whole can fulfill the objectives needed for the research.	☐	☐	☐	☐

Indicators	4	3	2	1
Objective				
8. Each item question/indicator requires only one specific answer or measures only one behavior.	☐	☐	☐	☐
Scale and Evaluation of Rating				
9. The scale adapted/used is appropriate for the item/indicator.	☐	☐	☐	☐

Remarks: *(recommendations for the improvement of the tool)*

_____

_____

_____

_____

_____

_____

Evaluator's Signature Over Printed Name

Adapted from:

Samosa, R. C. (2021). From Test Validity to Construct Validity & Back: Theory and Application in the Experimental Design in Science Action Research. Philippine Association of Physics and Science Instructors, De La Salle University, Manila.

APPENDIX E:
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CURRICULUM VITAE

Insert formal picture here

MARK DANZEN Q. AQUINO
Address: [Address]
Contact No.: [Contact No.]
Email Address: [Email Address]

EDUCATIONAL BACKGROUND:

TERTIARY:
- Bachelor's Degree: Bachelor of Science in Information Technology
- Institution Name: Colegio de San Gabriel Arcangel, Inc.
- Expected Year of Graduation: ____

SENIOR HIGH SCHOOL:
- Track/Strand:
- Institution Name:
- Year Graduated:
- Honors / Awards:

JUNIOR HIGH SCHOOL:
- Institution Name:
- Year Graduated:
- Honors / Awards:

PRIMARY:
- Institution Name:
- Year Graduated:

WORK EXPERIENCE:

Company Name:
Designation:
Address:
Inclusive Dates:

PERSONAL INFORMATION:

Age:
Sex:
Date of Birth:
Place of Birth:
Civil Status:
Religion:

I hereby certify that all information stated above is true and correct.

Mark Danzen Q. Aquino
Signature over printed name

CURRICULUM VITAE

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DARELL F. BACANI
Address: [Address]
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TERTIARY:
- Bachelor's Degree: Bachelor of Science in Information Technology
- Institution Name: Colegio de San Gabriel Arcangel, Inc.
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PERSONAL INFORMATION:

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Date of Birth:
Place of Birth:
Civil Status:
Religion:

I hereby certify that all information stated above is true and correct.

Darell F. Bacani
Signature over printed name

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CHRISTIAN GOLD DC. ESPAÑOLA
Address: [Address]
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EDUCATIONAL BACKGROUND:

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- Bachelor's Degree: Bachelor of Science in Information Technology
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- Expected Year of Graduation: ____

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Date of Birth:
Place of Birth:
Civil Status:
Religion:

I hereby certify that all information stated above is true and correct.

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Bachelor's Degree: Bachelor of Science in Information Technology
Institution Name: Colegio de San Gabriel Arcangel, Inc.
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Place of Birth:
Civil Status:
Religion:

I hereby certify that all information stated above is true and correct.

Markgil L. Gamboa
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Place of Birth:
Civil Status:
Religion:

I hereby certify that all information stated above is true and correct.

Irish I. Gregorio
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NICOLE S. MIJOY
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Religion:

I hereby certify that all information stated above is true and correct.

Nicole S. Mijoy
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EMMANUEL A. SPICER
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EDUCATIONAL BACKGROUND:

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I hereby certify that all information stated above is true and correct.

Emmanuel A. Spicer
Signature over printed name